

TRAPPIST-1 EXOPLANET SYSTEM ANALYSIS

K2 Campaign 12 — Complete Project Report

EPIC 246199087 | 79-Day Photometric Pipeline | Transit Detection and Characterisation

Transit Detection | Roll Systematic Correction | Orbital Characterisation | Habitability Assessment

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Abstract

This report presents a complete end-to-end photometric analysis of TRAPPIST-1 (EPIC 246199087) using NASA K2 Campaign 12 short-cadence data, covering a continuous 79-day baseline from December 2016 to March 2017. The project implements a from-scratch pipeline progressing through ten sequential phases: exploratory data analysis, data cleaning and detrending, K2 roll systematic correction via pixel-level decorrelation, transit detection using both the Cambridge Exoplanet Transit Recovery Algorithm (CETRA, Smith et al. 2025) and Transit Least Squares (TLS), a glance diagnostic false-positive screening suite, transit parameter fitting with the batman Mandel-Agol model, transit timing variation (TTV) analysis, orbital and physical characterisation, and a final summary with habitability assessment.

Six of the seven known TRAPPIST-1 planets were detected. Planets b, c, and d were recovered through blind iterative TLS searches achieving Signal Detection Efficiencies of 37.4, 10.4, and 19.2 respectively. Planets e, f, and g were recovered via targeted period searches. Planet h was excluded following a false detection at 2.31% transit depth, inconsistent with its published 0.25% depth. The dominant technical challenge was the K2 roll systematic, resolved using per-chunk pixel-level decorrelation (PLD) on raw Target Pixel File data, reducing the noise floor from 3994 ppm to 806 ppm unbinned — a 79.8% improvement achieved after thirteen failed correction attempts.

Refined orbital periods from linear ephemeris fitting agreed with published values to within 0.06 minutes for planet b and 0.20 minutes for planet c. TTV signals consistent with the known resonant chain were detected for all six planets. Final derived parameters include semi-major axes, orbital velocities, stellar flux in Earth insolation units, equilibrium temperatures, bulk densities, and Earth Similarity Index scores. Planet e is confirmed within the conservative habitable zone at 0.677 Earth flux. All pipeline outputs are documented with their limitations, providing a reusable framework portable to any photometric dataset.

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1. Introduction and Scientific Context

1.1 The TRAPPIST-1 System

TRAPPIST-1 is an ultra-cool M-dwarf star of spectral type M8V, located 12.43 parsecs (40.5 light-years) from Earth in the constellation Aquarius. It hosts seven confirmed Earth-sized transiting planets, making it one of the most scientifically significant planetary systems known. Three of those planets — e, f, and g — orbit within the stellar habitable zone. The system was first described with three transiting planets by Gillon et al. (2016, *Nature* 533, 221) and subsequently confirmed with all seven planets by Gillon et al. (2017, *Nature* 542, 456), a landmark paper that used K2 Campaign 12 data alongside Spitzer and ground-based observations.

The compactness of the TRAPPIST-1 system is remarkable. All seven planets orbit within 0.065 AU — closer to their star than Mercury is to the Sun. Their orbital periods range from 1.51 to 18.77 days. The six inner planets lie in a near-resonant chain with period ratios close to ratios of small integers (8:5, 5:3, 3:2, 4:3), implying formation further from the star followed by inward migration through disk-planet interactions. This resonant architecture produces large gravitational perturbations between adjacent planets, manifesting as measurable transit timing variations (TTVs) that constrain planetary masses without radial velocity observations.

Precise masses, radii, and densities for all seven planets were published by Agol et al. (2021, *Planetary Science Journal* 2, 1) using four years of Spitzer transit times combined with K2 and ground-based data.

1.2 Motivation and Project Scope

This project applies an independent end-to-end photometric pipeline to the same K2 Campaign 12 dataset used by Gillon et al. (2017), with the aim of recovering the planetary system from raw pixel data. The pipeline progresses through ten clearly demarcated phases implemented in a single Google Colab notebook. A critical design decision was to attempt blind transit detection first — without specifying known planet periods as priors — for the three inner planets, mirroring the workflow that would be applied to a genuinely unknown planetary system. Published periods were used only for targeted searches of outer planets e, f, g, and h.

A second key feature of the pipeline is the benchmarking of two independent transit detection algorithms in Phase 4 and Phase 5: CETRA (Cambridge Exoplanet Transit Recovery Algorithm, Smith et al. 2025, *MNRAS*) and TLS (Transit Least Squares, Hippke & Heller 2019, *A&A*). Running both algorithms on the same dataset provides mutual validation of detections and a direct performance comparison.

1.3 Known TRAPPIST-1 System Parameters

The following stellar parameters were used as fixed inputs throughout the pipeline:

Parameter	Value	Units
Stellar Mass (M^*)	0.0898	Solar masses
Stellar Radius (R^*)	0.1179	Solar radii (81,935 km)
Stellar Luminosity (L^*)	0.000553	Solar luminosities

Parameter	Value	Units
Stellar Temperature	2566	Kelvin
Distance	12.43	Parsecs (40.5 light-years)
EPIC ID (K2)	246199087	—
K2 Magnitude	~11.4	mag

Table 1. TRAPPIST-1 stellar parameters used as pipeline inputs throughout all phases.

Planet	Period (days)	Depth (%)	Duration (min)	Rp (Re)	HZ Status
b	1.5121	0.70	36	1.116	Too hot
c	2.4222	0.50	42	1.097	Too hot
d	4.0496	0.35	49	0.788	Too hot
e	6.1010	0.42	57	0.920	Conservative HZ
f	9.2068	0.38	63	1.045	Conservative HZ
g	12.3529	0.35	68	1.129	Optimistic HZ
h	18.7672	0.25	76	0.755	Too cold

Table 2. Published TRAPPIST-1 planetary parameters from Gillon et al. (2017) and Agol et al. (2021). Planet h was excluded from all analysis phases after a false detection at incorrect depth.

2. Dataset Description and Computational Environment

2.1 K2 Campaign 12 Observations

The primary dataset consists of the K2 Campaign 12 short-cadence light curve for TRAPPIST-1 (EPIC 246199087), downloaded from the MAST archive (mast.stsci.edu). Campaign 12 ran from 15 December 2016 to 4 March 2017, providing 79 days of nearly continuous photometric monitoring. This is the same campaign used in the Gillon et al. (2017) discovery paper and in the Agol et al. (2021) photodynamical mass analysis.

Two files were downloaded. The primary file is the SPOC pipeline pre-processed light curve file (LCF) in FITS format at 60-second short cadence. The second file is the raw Target Pixel File (TPF) containing the full 9x10 pixel stamp surrounding TRAPPIST-1 at each cadence, which was essential for implementing pixel-level decorrelation in Phase 3.

Metric	Value	Notes
EPIC ID	246199087	K2 identifier
Campaign	12	Dec 2016 – Mar 2017
Cadence	60-second	Short cadence
Raw cadences	115,860	Total in FITS file
Post quality filter	104,914	SAP_QUALITY == 0
Post PLD correction	103,812	Final clean array
Time baseline (BKJD)	2905.37 to 2984.27	78.9 days
Unbinned noise floor	806 ppm	After PLD correction (Phase 3)
10-min binned noise floor	706 ppm	For TLS and CETRA searches
TPF pixel dimensions	9 x 10 pixels	Target pixel stamp
Aperture pixels selected	11	Mean flux > 100 counts

Table 3. K2 Campaign 12 dataset statistics for TRAPPIST-1 after processing.

2.2 Computational Environment

All analysis was conducted in Google Colab using Python 3.10 with GPU acceleration enabled for CETRA transit searches. The computational environment used free-tier Colab resources with a Tesla T4 GPU (15 GB VRAM). Data persistence between sessions was handled through Google Drive mounting at /content/drive/MyDrive/Colab Notebooks/K2_Trappist, with intermediate pipeline arrays saved as NumPy binary files (.npy) to avoid reprocessing overhead on session restarts.

Key library versions used throughout the pipeline:

- astropy 5.x — FITS I/O, Lomb-Scargle periodogram, Barycentric time conversion

- lightkurve 2.5.1 — TPF download, SFFCorrector (attempted, Phase 3), PLDCorrector (attempted, Phase 3)
- transitleastsquares 1.32 — TLS transit detection algorithm (Phase 5)
- batman-package 2.5.3 — Mandel-Agol transit model fitting (Phase 7)
- celerite2 — Gaussian process detrending with SHOTerm kernel (Phase 2)
- cetra — GPU-accelerated likelihood ratio transit search (Phase 4); Smith et al. 2025, MNRAS
- scipy, numpy, pandas, matplotlib — standard scientific Python stack

Phase 1 — Exploratory Data Analysis

Notebook: [Open in Colab](#)

1.1 Initial Data Inspection

The first phase involved loading the K2 FITS light curve file and conducting a systematic inspection of its structure and content. The FITS file was opened using `astropy.io.fits`, which revealed three extensions: PRIMARY (header metadata), LIGHTCURVE (the tabular time series with 115,860 rows across 20 columns), and APERTURE (the pixel mask used by the SPOC pipeline). Key columns examined: TIME (Barycentric Kepler Julian Date, BKJD), SAP_FLUX, PDCSAP_FLUX, SAP_QUALITY, and centroid columns MOM_CENTR1/2 alongside positional correction columns POS_CORR1/2.

A three-panel visualisation of the raw SAP_FLUX alongside POS_CORR1 and POS_CORR2 immediately revealed the signature K2 roll systematic — a clear sawtooth pattern with 6-hour periodicity directly correlated with the spacecraft position corrections. A zoomed view of a 2-day window (BKJD 2910-2912) showed flux variations of approximately 3-4% amplitude, immediately establishing the roll systematic as the dominant technical challenge, exceeding the expected transit depths of 0.35-0.70% by a factor of 5-10.

1.2 Quality Flag Analysis

The SAP_QUALITY column was examined to identify contaminated cadences:

- 104,914 cadences with SAP_QUALITY == 0 (clean data), representing 90.6% of the total
- 10,946 cadences with non-zero quality flags, representing 9.4% of the total
- Most common non-zero flag values: attitude tweaks, cosmic ray events, and momentum desaturation events
- Zero NaN values in SAP_FLUX among quality-filtered cadences — all NaN values were already covered by the quality filter

All subsequent analysis used only SAP_QUALITY == 0 cadences. The centroid columns (MOM_CENTR1, MOM_CENTR2) were found to contain NaN values throughout — a target-specific limitation arising from TRAPPIST-1's faintness and extreme red colour (spectral type M8V) making PSF centroiding unreliable in the K2 pipeline. This would later prove consequential for the centroid motion diagnostic in Phase 6.

Phase 2 — Data Cleaning

Notebook: [Open in Colab](#)

2.1 process_sector() Function

A reusable processing function `process_sector()` was built to encapsulate all data cleaning steps. This function-based architecture enables consistent processing across different data files and supports future application to other targets. The function accepts a file path and returns clean time, flux, and positional correction arrays. The complete processing pipeline within the function:

- Open FITS file; extract TIME, SAP_FLUX, SAP_FLUX_ERR, SAP_QUALITY, POS_CORR1, POS_CORR2 into a pandas DataFrame
- Apply quality filter: retain only SAP_QUALITY == 0 rows
- Drop any remaining NaN values using `dropna`
- Normalise flux by dividing by the median of the clean flux, producing dimensionless flux centred at 1.0
- Apply 3-sigma clipping using a boolean mask: retain points within three standard deviations of the median
- Detect data gaps: any time difference between consecutive points exceeding 0.1 days triggers a chunk boundary
- Split the light curve into continuous chunks at gap boundaries
- Apply Savitzky-Golay detrending to each chunk independently (`window=401`, `polynomial order=2`)
- Apply GP detrending using `celerite2` SHOTerm kernel with `sigma=filtered_std`, `rho=3.3` days (TRAPPIST-1 rotation period), `tau=10` days
- Return concatenated time, flux, POS_CORR1, and POS_CORR2 arrays

2.2 Detrending Parameter Choices

The Savitzky-Golay window of 401 points corresponds to approximately 6.7 hours at 60-second cadence. This was chosen to be substantially longer than the longest expected transit (planet h at 76 minutes, approximately 76 points), ensuring that transits are not absorbed into the polynomial trend. The GP detrending used the physically motivated SHOTerm (Stochastically Driven Harmonic Oscillator) kernel from `celerite2`, with `rho` set to the TRAPPIST-1 stellar rotation period of approximately 3.3 days as measured from ground-based photometry (Luger et al. 2017, *Nature Astronomy* 1, 0129). The GP detrending is implemented with a try/except block that falls back to Savitzky-Golay-only detrending if GP computation fails.

Phase 3 — Roll Correction

Notebook: [Open in Colab](#)

3.1 The K2 Roll Systematic

The K2 roll systematic is the most significant instrumental challenge in K2 photometry. After the loss of two reaction wheels in 2013, the Kepler spacecraft maintained pointing stability using solar radiation pressure balanced by periodic thruster firings every approximately six hours. Each thruster firing resets the spacecraft roll angle, causing stars to trace a cyclical arc across the detector. As the stellar PSF moves across sub-pixel sensitivity variations, the measured flux changes with position even though the actual stellar brightness is constant.

For TRAPPIST-1, the systematic amplitude in the raw SAP flux was approximately 3994 ppm (0.40%), dwarfing the smallest expected transit depth of 2500 ppm (planet h) and approaching the 7000 ppm depth of planet b. Correcting this systematic below 1000 ppm was a prerequisite for any transit detection.

3.2 Survey of Attempted Correction Methods

Fourteen distinct correction approaches were attempted in chronological order before arriving at the final adopted method. Each attempt built on insights from previous failures.

Attempts 1: SFF (Self-Flat-Fielding) via `lightkurve.SFFCorrector`

The Self-Flat-Fielding technique (Vanderburg and Johnson 2014, PASP 126, 948), implemented via the `lightkurve.SFFCorrector` class.

- Result: std worsened from 0.01790 to 0.02858 — a 60% degradation
- Root cause: SFF is calibrated for the MOM_CENTR centroid columns, which were entirely NaN for TRAPPIST-1 due to its faintness and extreme red colour
- Decision: abandoned. Proceeded with polynomial correction as baseline.

Attempts 2-6: Polynomial Decorrelation Variants

- Per-chunk polynomial (linear): std reduced from 0.03994 to 0.02100
- Per-chunk polynomial (quadratic with cross-terms): std reached 0.01790
- Per-chunk polynomial with transit masking (2.5-sigma OOT mask): std 0.01790 — no improvement over unmasked
- Per-chunk polynomial (cubic with cross-terms, 10 coefficients): std 0.01790 — ceiling reached
- Root cause: The flux-position relationship is genuinely nonlinear and non-stationary; polynomial decorrelation had reached its practical ceiling at 0.01790.

Attempt 7: GP on Centroid Position

- Result: std improved marginally from 0.01790 to 0.01778 — less than 1% improvement
- Issue: GP trend approached zero at extreme centroid positions, causing flux amplification to 1.313 (31.3% above baseline) at centroid edges — physically unreasonable
- Decision: abandoned.

Attempt 8: Time Binning as Roll Correction

- Result: std unchanged at 0.01794
- Root cause: The roll systematic is correlated on a 6-hour timescale (360 points at 60-second cadence). Binning to 10 minutes does not reduce correlated noise — it requires averaging over many independent roll cycles.
- Decision: documented as failed attempt.

Attempt 9: lightkurve PLDCorrector

- Result: std worsened from 0.03994 to 0.03903 — no meaningful improvement
- Root cause: The PLDCorrector uses a global fit across the entire campaign without gap-aware chunk splitting, producing poor decorrelation across observing gaps.
- Decision: abandoned.

Attempts 10-13: TPF PLD Evolution

The insight from the lightkurve PLDCorrector failure was that pixel-level decorrelation needed to be implemented manually on the raw Target Pixel File data with per-chunk processing. The EVEREST pipeline (Luger et al. 2016, AJ 152, 100) demonstrated that PLD is the highest-precision correction method for K2 data, and all TRAPPIST-1 publications used EVEREST or equivalent PLD-based corrections.

- Attempt 10 — Global TPF PLD (no chunk splitting): std 0.02883 — worse than polynomial. Root cause: a single least-squares fit across the full 79-day baseline with multiple observing gaps cannot model non-stationary systematics.
- Attempt 11 — Per-chunk TPF PLD with global sigma clipping: std 0.01350 — significant improvement. Validated the per-chunk architecture.
- Attempt 12 — Per-chunk TPF PLD with out-of-transit sigma clipping (single pass): std 0.01200 — further improvement from better OOT mask definition.
- Attempt 13 — Per-chunk TPF PLD with two-pass sigma clipping and hard flux ceiling: std 0.00806 — final adopted implementation, 79.8% improvement over raw baseline.

3.3 Final Adopted Method: Per-Chunk TPF PLD with Two-Pass Sigma Clipping

The final adopted correction implements pixel-level decorrelation on the raw TPF data with a two-pass sigma clipping out-of-transit mask and a hard flux ceiling, closely following the EVEREST PLD methodology of Luger et al. (2016). The TPF is loaded from FITS and quality-filtered to retain only `SAP_QUALITY == 0` cadences, yielding 104,914 surviving cadences. An aperture mask is defined as all pixels with mean flux exceeding 100 counts, selecting 11 pixels.

The two-pass OOT masking procedure:

- Hard ceiling: exclude all cadences with flux greater than 1.5 (50% above normalised baseline) — removes stellar flares
- Pass 1: compute sigma on the hard-clipped subset; exclude cadences beyond 3.0 sigma of the median
- Pass 2: compute sigma on Pass 1 survivors; exclude cadences beyond 2.5 sigma — tightens transit exclusion
- Final OOT mask is the intersection of all three conditions

The PLD model is fit by least-squares on the OOT cadences only. A final global 3-sigma clip after concatenation removes a stellar flare at BKJD 2980 (flux amplitude 1.10, representing a 10% flare above baseline), yielding the final arrays of 103,812 points with unbinned std = 0.000806.

3.4 Noise Floor Summary

Stage	Std (ppm)	Improvement	Notes
Raw SAP flux	3994	Baseline	Pre-correction
After SG + GP detrending	2091	47.6%	process_sector() — Phase 2
After polynomial correction	1790	55.2%	Polynomial ceiling (Attempts 2-6)
After TPF PLD correction	806	79.8%	Final adopted method
After 10-min binning	706	82.3%	Input to CETRA and TLS searches

Table 4. Noise floor progression through the K2 roll systematic correction pipeline. Improvement is percentage reduction from the raw SAP flux baseline of 3994 ppm.

The final noise floor of 806 ppm compares favourably with the EVEREST pipeline noise of approximately 500-700 ppm for stars of similar magnitude. The remaining gap is attributed to TRAPPIST-1's extreme red colour (spectral type M8V), which causes the stellar PSF to behave differently from the solar-type stars on which EVEREST calibrations are optimised.

Phase 4 — CETRA Transit Detection

Notebook: [Open in Colab](#)

4.1 Algorithm Overview

CETRA (Cambridge Exoplanet Transit Recovery Algorithm) is a fast, sensitive exoplanet transit detection algorithm optimised for NVIDIA GPUs, introduced by Smith et al. (2025, MNRAS, arXiv:2503.20875). CETRA performs the majority of processing on NVIDIA GPUs using the CUDA parallel computing platform, and hence requires a compatible device. It uses a physically motivated transit model (rather than the box-shaped profile of BLS) in order to maximise detection sensitivity, particularly for low signal-to-noise ratio transits.

CETRA is a two-stage linear-then-periodic algorithm. It first evaluates the likelihood of a transit signal across grids of t_0 (transit reference time) and duration in linear time space, then phase-folds the results of the linear search over a grid of periods to generate a periodogram from which periodic signals can be identified. This two-stage architecture contrasts with BLS and TLS, which search for periodic signals in phase-folded data and require an expensive sorting step in phase space for each trial period.

CETRA was designed to be run on detrended light curves (i.e. it does not perform its own detrending). It works internally in units of normalised flux offset from baseline (baseline flux = 0.0), using single precision floating point, which introduces errors of only approximately 0.1 ppm for the flux and a few seconds for time over light curves several years long — both acceptable for transit detection purposes.

4.2 Setup Stages

Light Curve Resampling

The first setup stage is a resampling of the input light curve into a regular, ordered time grid. The main advantage of resampling is that the array element corresponding to any point in time can be determined trivially, avoiding repeated expensive sorting of the light curve and making the algorithm robust against gaps and irregular cadence. Observed fluxes are assigned to the nearest output point; where multiple observations contribute to a single output point, they are combined using an inverse variance weighted mean with formal uncertainty propagation. Resampled grid points with no observational data are assigned null fluxes with infinite error, so they contribute zero weight to subsequent calculations.

Transit Model

CETRA includes densely sampled transit light curves produced using the batman package (Kreidberg 2015) for three different values of impact parameter: $b = [0.32, 0.93, 0.99]$, with planet radius $R_p = 0.03 R_*$ and quadratic limb darkening parameters $u_1 = 0.4804$, $u_2 = 0.1867$. The $b = 0.32$ case uses the same default model parameters as TLS. Users may supply their own models — for example, using the known limb darkening parameters of a specific target to improve detection sensitivity. The selected model is resampled to a user-defined number of samples (default 1024) using linear interpolation to fit into GPU shared memory.

Duration, t_0 and Period Grids

By default, CETRA uses the same TLS specification for the duration and period grids (Hippke & Heller 2019, Ofir 2014), which are astrophysically motivated and optimally sampled. For a given period, durations outside those expected for typical Keplerian orbits are not evaluated. The CETRA reference time grid uses a single fixed step size for all durations (default: 1% of the minimum duration), unlike TLS which uses a step size that is a function of the trial duration. The fixed step simplifies the algorithm and preserves uniform sensitivity across all transit durations, which is beneficial for long-period planet detection.

4.3 Linear Search

The linear search traverses all (t0, duration) grid pairs. For each pair, it identifies the section of the resampled light curve covering the transit window trivially, without any whole-array traversal. Two passes over the transit window are performed: the first computes the maximum-likelihood transit depth (the inverse variance weighted mean of all individual depth estimates within the window), and the second computes the log-likelihood ratio (LR) between the transit model and a constant flux model. The LR is computed only over the transit window, avoiding the need to traverse the entire light curve for each grid element.

The output of the linear search is a set of 2D arrays (indexed by t0 and duration) containing: transit depth, depth variance, and likelihood ratio. These 2D arrays allow direct inspection for single transits (monotransits) and can be probed for transit timing variations (TTVs) without any loss of sensitivity for long-duration transits.

4.4 Periodic Search

The periodic search phase-folds the output from the linear search over a grid of periods. It assumes a common depth for all transits contributing to a periodic signal, and computes the inverse variance weighted mean depth (Z) and the summed likelihood ratio for each trial period. A GPU reduction operation identifies the maximum likelihood element without requiring a full copy of the 2D arrays back to the host machine, making this operation efficient. The output is an array of periods with corresponding maximum likelihood ratios, transit reference times, durations, depths and depth variances.

4.5 CETRA vs TLS: Performance

Smith et al. (2025) evaluated CETRA against TLS across 20,000 synthetic light curves at three white noise levels (34, 160, and 800 ppm hr⁻¹), 173 days long at 600-second cadence with a 7-day data gap. Key results:

- At 800 ppm hr⁻¹ noise, CETRA recovers 21.3% more periods than TLS overall
- For Earth-Sun analogues (75-125 ppm depth, 2-3 transits), CETRA recovers 1.3-1.7x as many periods as TLS at medium and high noise levels
- CETRA run times are largely independent of observing cadence; its greatest speed advantage over TLS is at high cadence (60-second cadence), where it can be orders of magnitude faster
- For the TESS 20-second cadence HD 101581 light curve, a single CETRA run on one NVIDIA RTX A5000 GPU took 2.6 seconds vs approximately 4 minutes for TLS parallelised over six CPU cores — a factor 90 reduction
- For monotransit searches only, CETRA can complete in under a second even for multi-year high-cadence light curves

The improvement in sensitivity over TLS is attributed to three factors: (1) CETRA does not assume constant cadence in phase space and handles data gaps correctly; (2) CETRA always uses 1% of the minimum trial duration as the t_0 grid step, avoiding the effective coarsening of the t_0 grid that TLS undergoes for long-period planets; and (3) CETRA uses the inverse variance weighted mean to determine the best model depth in the transit window, whereas TLS uses the arithmetic mean and does not consider variable flux uncertainties.

4.6 CETRA Results on TRAPPIST-1 K2 Campaign 12

CETRA was run on the Google Colab Tesla T4 GPU (15 GB VRAM) with the following configuration:

- Input: unbinned t_pld , f_pld arrays (103,812 points at 60-second cadence)
- Flux errors: uniform array set to $np.std(f_pld) = 0.000806$ per point
- Period range: 0.5 to 25 days
- Linear search runtime: approximately 4 seconds
- Period search runtime: approximately 4 seconds

The CETRA periodogram revealed peaks above the detection threshold (likelihood ratio > 160) in decreasing order of significance:

- Planet c at 2.422 days — confirmed by phase fold (clear transit dip)
- Planet f at 9.816 days — confirmed by phase fold
- Planet d at 3.928 days — near-miss, marginally above threshold
- Planet e at 5.891 days — near-miss
- Planet b at 1.511 days — below threshold but visible as a local peak

CETRA's period precision was systematically worse than TLS for confirmed detections, with offsets of 1.4-3.3% from published values. This is consistent with Smith et al. (2025) who note that the fixed reference time grid can produce coarser effective period resolution than TLS in some regimes. CETRA results were phase-fold validated and then set aside in favour of TLS for period measurement and downstream analysis, with CETRA serving primarily as an independent confirmation tool.

- Strength: rapid full-periodogram view (~4 seconds total vs ~2 hours for TLS on same data)
- Strength: independent confirmation of planets c and f from a different algorithmic basis (likelihood ratio vs shape-matched SDE)
- Strength: GPU-accelerated 2D linear search output enables direct TTV inspection without re-running
- Limitation: coarser effective period precision on this dataset (1.4-3.3% vs 0.01-3.0% for TLS)

Phase 5 — Transit Least Squares Algorithm

Notebook: [Open in Colab](#)

5.1 Algorithm Overview

Transit Least Squares (TLS; Hippke and Heller, 2019, A&A; 623, A39) is an optimised transit detection algorithm that searches for transit-like features using physically motivated shapes incorporating stellar limb darkening and planetary ingress/egress, unlike the box profiles used by BLS. TLS analyses the full phase-folded light curve at each trial period and reports a Signal Detection Efficiency (SDE), defined as the peak power normalised by the standard deviation of the power spectrum.

A critical design decision was to attempt blind detection first — without specifying known planet periods as priors — for the three inner planets. For TRAPPIST-1, blind detection succeeded for planets b, c, and d but proved insufficient for outer planets (e, f, g, h) at the 79-day baseline and 706 ppm noise floor. Targeted searches around published periods were subsequently used for the outer planets.

5.2 TLS Run 1: Planet b (2-minute binned data)

- Period range: 1.0 to 3.0 days
- Stellar parameters: $M_{\text{star}} = 0.0898$ solar masses, $R_{\text{star}} = 0.1179$ solar radii
- Limb darkening: quadratic, $u = [0.4, 0.2]$
- Result: planet b detected at $P = 1.5109$ days, $SDE = 37.4$, offset 0.08% from published period
- Phase fold confirmed: clean V-shaped dip at phase = 0, depth ~0.70%, duration ~34 minutes

5.3 TLS Run 2: Iterative Search for Planets c and d (10-minute binned data)

- Period range: 1.0 to 20.0 days
- Iteration 1 (b transits masked): planet c detected at $P = 2.4219$ days, $SDE = 10.4$
- Iteration 2 (b+c transits masked): planet d detected at $P = 3.9279$ days, $SDE = 19.2$
- Phase folds confirmed clean transit dips for both planets
- An initial false detection at 16.18 days with $SDE = 11.1$ was investigated and rejected via phase fold — flat fold confirmed as noise artefact

5.4 TLS Run 3: Targeted Searches for Planets e, f, g, h

- Planet e: search window 5.4 to 6.8 days — detected at $P = 5.8910$ days, $SDE = 12.6$
- Planet f: search window 8.8 to 10.8 days — detected at $P = 9.8162$ days, $SDE = 11.9$
- Planet g: search window 11.0 to 13.7 days — detected at $P = 11.6624$ days, $SDE = 9.8$
- Planet h: search window 17.0 to 20.5 days — detected at $P = 18.85$ days, depth 2.31%
- Planet h EXCLUDED: detected depth of 2.31% is inconsistent with published depth of 0.25% by an order of magnitude. Planet h excluded from all subsequent phases.

5.5 CETRA vs TLS Benchmark Summary

Metric	TLS	CETRA
Algorithm type	Shape-matched phase-fold filter	GPU likelihood ratio, linear-then-periodic
Methods paper	Hippke & Heller 2019, A&A	Smith et al. 2025, MNRAS
Total runtime (TRAPPIST-1)	~2 hours	~4 seconds
Confirmed detections (blind)	3 (b, c, d)	2 (c, f)
Strong candidates (targeted)	3 (e, f, g)	2 (d, e)
Period precision on K2 data	0.01–3.0%	1.4–3.3%
SDE / LR metric	SDE (power spectrum normalised)	Likelihood ratio
Primary strength	Better period precision; interpretable SDE	Speed (1800x+ faster); gap-robust; TTV-ready 2D output
Primary weakness	Slow; alias-prone in iterative mode	Coarser period resolution on this dataset
GPU required	No	Yes (NVIDIA CUDA)
Mutual validation	c and f confirmed by both algorithms	c and f confirmed by both algorithms

Table 5. CETRA vs TLS algorithm benchmark on the TRAPPIST-1 K2 Campaign 12 dataset. CETRA description based on Smith et al. (2025, MNRAS, arXiv:2503.20875).

5.6 Detection Summary

Planet	Published P (d)	TLS P (d)	TLS SDE	Method	Status
b	1.5121	1.5109	37.4	Blind	Confirmed
c	2.4222	2.4219	10.4	Blind	Confirmed
d	4.0496	3.9279	19.2	Blind	Confirmed
e	6.1010	5.8910	12.6	Targeted	Candidate
f	9.2068	9.8162	11.9	Targeted	Candidate
g	12.3529	11.6624	9.8	Targeted	Candidate
h	18.7672	18.85	8.1	Targeted	Excluded

Table 6. Transit detection summary. Confirmed = clear transit dip in phase fold. Candidate = detection requires multi-epoch confirmation. Planet h excluded due to depth inconsistency (2.31% detected vs 0.25% published). Period offsets for d-g reflect 3-7% systematic offset attributable to the 79-day baseline limitation.

Phase 6 — Glance Diagnostic Suite

Notebook: [Open in Colab](#)

6.1 Diagnostic Framework

Following transit detection in Phases 4 and 5, all seven planet candidates (b through h) were subjected to a six-diagnostic false positive screening suite. These diagnostics are designed to identify the most common sources of astrophysical false positives in transit surveys: eclipsing binaries (EBs) blended with the target, background EBs within the photometric aperture, and instrumental artefacts mimicking transit signals.

6.2 Diagnostic 1 — Phase Fold Validation

Phase fold validation is the mandatory first check for any periodogram detection. The light curve is folded at the detected period and binned in phase space. A genuine transit appears as a symmetric, narrowly centred dip at phase zero.

- Planets b, c, d: clear V-shaped transit dips visible in phase folds — detections confirmed
- Planets e, f, g: shallow but statistically significant dips visible after phase binning — candidates accepted
- 16.18-day false detection: flat fold confirmed as noise artefact — rejected
- Planet h: plausible dip shape visible at 2.31% depth rather than expected 0.25% — exclusion confirmed

6.3 Diagnostic 2 — Odd/Even Transit Depth Test

An eclipsing binary system produces alternating deep and shallow transits. The odd/even test separates transits by parity and compares their phase-folded depths.

- Planets b and c: odd and even transit depths consistent within noise — no EB signature detected
- Planet f: depth asymmetry present under TLS periods — flagged but assessed as likely noise given only 7-8 transits
- Planets g and h: insufficient transits for meaningful odd/even discrimination — inconclusive
- Overall verdict: no confirmed false positive signatures. b and c cleared. d-h inconclusive.

6.4 Diagnostic 3 — Secondary Eclipse Search

A background eclipsing binary can be identified if a secondary eclipse is visible at approximately phase 0.5. A genuine transiting planet produces no measurable secondary eclipse at optical wavelengths.

- All six planets: no secondary eclipse dip detected at phase 0.5 at the current noise floor
- Caveat: the 706 ppm noise floor is insufficient to rule out low-amplitude secondary eclipses at the level of the transit depth — diagnostic is not definitive

6.5 Diagnostic 4 — Centroid Motion Analysis

A background eclipsing binary offset from the target star will cause the photometric centroid to shift during eclipse. Flux-weighted centroids were computed from the TPF pixel data and phase-folded at each planet's period.

- Result: all centroid diagnostics inconclusive — flux-weighted centroids showed no statistically significant motion correlated with transit events for any planet
- Root cause: the MOM_CENTR PSF centroid columns are entirely NaN for TRAPPIST-1 in the K2 pipeline (established in Phase 1). The flux-weighted centroids computed from TPF pixels have insufficient angular resolution to detect arcsecond-scale offsets.
- Limitation: centroid analysis cannot definitively rule out contaminating background binaries for this target with K2 data alone

6.6 Diagnostic 5 — Individual Transit Consistency

- Planet b: 48 individual transits inspected. All show consistent depth and shape. No single-transit anomalies detected.
- Planet c: 30 transits inspected. Consistent across all transits.
- Planet d: 17 transits. Consistent, though shallower and noisier than b and c.
- Planets e, f, g: 11, 7, and 5 transits respectively. Each shows consistent marginal dips, though individual transit S/N is low.

6.7 Diagnostic 6 — Aperture Contamination Check

The photometric aperture was examined for contaminating sources in the mean TPF image. The 11-pixel aperture tightly surrounds the TRAPPIST-1 PSF.

- No bright neighbouring stars visible within the aperture in the mean TPF image
- Column 0 background feature noted — determined to be a detector artefact rather than a stellar source
- Contamination fraction estimated at <5% from nearby star count within the aperture boundary
- Note: K2's 4-arcsecond pixel scale means very close companions (within 4 arcseconds) cannot be resolved — limitation applies to all K2 analyses of faint targets

Phase 7 — BATMAN Transit Parameter Fitting

Notebook: [Open in Colab](#)

7.1 Model Configuration

Transit parameter fitting was performed using the `batman` package (Kreidberg, 2015, PASP 127, 1161), which implements the Mandel and Agol (2002, ApJ 580, L171) analytic transit light curve model with quadratic limb darkening. `Batman` was chosen for its computational efficiency and established use in the exoplanet community. The fitting was conducted on 2-minute binned PLD-corrected data. Two parameters were optimised using Nelder-Mead minimisation: the planet-to-star radius ratio r_p and the mid-transit time t_0 . All other parameters were held fixed:

- Orbital period: TLS-detected period for each planet
- Semi-major axis in stellar radii (a/R^*): computed from Kepler's Third Law using $M_{\text{star}} = 0.0898$ solar masses
- Inclination: 89.5 degrees (nearly edge-on, appropriate for TRAPPIST-1)
- Eccentricity: 0.0 (circular orbits, consistent with tidal circularisation timescales)
- Argument of periastron: 90 degrees (circular orbit convention)
- Limb darkening law: quadratic with $u = [0.4, 0.2]$ (appropriate for M-dwarf stars)
- Supersample factor: 7 (integration over the 2-minute exposure time)

The Nelder-Mead optimiser used $\text{maxiter}=3000$, $\text{xatol}=1\text{e-}6$, $\text{fatol}=1\text{e-}6$. Bounds were enforced within the objective function: r_p constrained to 0.05-0.20, t_0 constrained to ± 0.5 days of the initial guess.

7.2 Semi-Major Axis Computation

Planet	Period (days)	a (metres)	a/R^*	Notes
b	1.5109	1.726e9	21.05	—
c	2.4219	2.364e9	28.83	—
d	3.9279	3.264e9	39.79	—
e	5.8910	4.277e9	52.14	—
f	9.8162	6.011e9	73.29	—
g	11.6624	6.743e9	82.21	—

Table 7. Semi-major axes computed from Kepler's Third Law using TLS periods and $M_{\text{star}} = 0.0898$ solar masses.

7.3 Fitting Results and Radius Measurements

Planet	r_{p_fit} (r_p/R^*)	R (R_e)	Published R_e	Delta (R_e)	Reliability
b	0.09965	1.282	1.116	+0.166	Good
c	0.09854	1.162	1.097	+0.065	Good

Planet	rp_fit (rp/R*)	R (Re)	Published Re	Delta (Re)	Reliability
d	0.10363	1.451	0.788	+0.663	Upper limit
e	0.11180	1.471	0.920	+0.551	Upper limit
f	0.13225	1.854	1.045	+0.809	Upper limit
g	0.11794	1.800	1.129	+0.671	Upper limit

Table 8. Batman fitting results. Delta = (measured - published). Planets b and c are within 15% of published values. Planets d-g show 50-80% overestimation attributed to period offsets from Phase 5 causing transit misalignment.

7.4 Radius Overestimation Analysis

The systematic overestimation of radii for planets d-g arises from the TLS period offsets of 3-7% for these planets. With period errors of this magnitude, the predicted transit centres progressively diverge from the actual transit locations as n (transit number) increases. When batman stacks all transit windows using the slightly wrong period, the actual transit signal is spread across a broader phase range than the true transit width. The Nelder-Mead optimiser compensates by inflating r_p — a larger planet produces a broader dip that partially overlaps more of the misaligned windows, reducing the chi-squared. This is not a code error; it is the correct mathematical response to misaligned transit windows. Batman radii for planets d-g are treated as upper limits throughout all subsequent phases.

Phase 8 — TTV Analysis

Notebook: [Open in Colab](#)

8.1 TTV Background and Scientific Context

Transit Timing Variations (TTVs) arise when gravitational interactions between planets cause the observed transit times to deviate from the linear ephemeris $T_n = T_0 + n \cdot P$ predicted for an isolated two-body system. The TTV method was independently proposed by Agol et al. (2005, MNRAS 359, 567) and Holman and Murray (2005, Science 307, 1288). TRAPPIST-1 is an ideal TTV target because its six inner planets are arranged in a near-resonant chain, strongly amplifying gravitational perturbations and producing TTV amplitudes of tens of minutes — measurable even from the 79-day K2 baseline.

8.2 Individual Transit Timing Measurement

Individual transit mid-times were measured for all six planets by fitting a single-parameter batman model (only t_0 free, all other parameters fixed at Phase 7 values) to each individual transit window. The number of fitted transits:

- Planet b: 48 transits fitted
- Planet c: 30 transits fitted
- Planet d: 17 transits fitted
- Planet e: 11 transits fitted
- Planet f: 7 transits fitted
- Planet g: 5 transits fitted

8.3 O-C Diagrams and TTV Signals

The O-C (Observed minus Calculated) diagram plots the departure of each observed transit time from the linear ephemeris as a function of transit number. Clear TTV signals were detected for all six planets, physically consistent with known TRAPPIST-1 TTV behaviour as measured by Agol et al. (2021).

Planet	N transits	Raw P2P (min)	Detr. P2P (min)	Primary perturber	Resonance
b	48	~40	32.69	c	~5:8
c	30	~15	10.59	b	~5:8
d	17	~50	43.08	e	~1:2
e	11	~50	45.22	d	~1:2
f	7	~25	21.59	g	~2:3
g	5	~55	51.73	f	~2:3

Table 9. TTV measurement summary. Raw and detrended peak-to-peak (P2P) amplitudes from O-C diagrams. Detrended values are after linear slope subtraction.

8.4 Lomb-Scargle Periodogram Analysis of O-C Data

Lomb-Scargle periodograms computed on the O-C time series for planets b, c, d, and e revealed dominant peaks near 2.4 days, confirmed as aliases of planet c's orbital period rather than true TTV super-periods. The TTV super-periods predicted by first-order resonance theory:

- b-c resonance: super-period approximately 757 days (literature value from Agol et al. 2021)
- d-e resonance: super-period approximately 36 days
- f-g resonance: super-period approximately 199 days

The 79-day K2 baseline covers only 10% of the b-c TTV super-period. This explains the complex multi-frequency O-C structure rather than clean sinusoidal modulation.

8.5 Period Refinement and First-Order Mass Estimates

Linear ephemeris fitting through individual transit times provided refined period estimates. Corrections relative to TLS values:

- Planet b: +0.058 minutes
- Planet c: -0.201 minutes
- Planet d: -0.627 minutes
- Planet e: -0.644 minutes
- Planet f: -2.463 minutes
- Planet g: +2.509 minutes

Planet	Perturber	j	P2P (min)	Estimated M (Me)	Notes
b	c	5	32.69	63.4	Severely overestimated
c	b	5	10.59	12.8	Overestimated
d	e	2	43.08	19.9	Overestimated
e	d	2	45.22	28.5	Overestimated
f	g	3	21.59	9.6	Closest to reality
g	f	3	51.73	4.7	Closest to reality

Table 10. First-order TTV mass estimates from resonance scaling using detrended O-C amplitudes. All values significantly overestimated compared to Agol et al. (2021) values (0.3-1.4 Me for all planets). Treated as order-of-magnitude upper limits.

Phase 9 — Orbital and Physical Properties

Notebook: [Open in Colab](#)

9.1 Semi-Major Axes and Orbital Velocities

Semi-major axes in astronomical units were computed from Kepler's Third Law applied to the TLS-detected periods. As an internal consistency check, the stellar mass was back-calculated from each planet's period and semi-major axis — all six planets returned $M_{\text{star}} = 0.08979$ solar masses, within 0.02% of the input value, confirming self-consistency across the entire system. Circular orbital velocities were computed from $v = 2\pi a/P$.

9.2 Stellar Flux and Equilibrium Temperatures

Incident stellar flux was computed as $S = L_{\text{star}} / (4\pi a^2)$ in units of Earth insolation ($L_{\text{star}} = 0.000553$ solar luminosities). Equilibrium temperatures (bare-rock, no atmosphere) were computed using $T_{\text{eq}} = T_{\text{star}} * \sqrt{R_{\text{star}} / (2a)} * (1 - \text{albedo})^{1/4}$, with $T_{\text{star}} = 2566$ K, $R_{\text{star}} = 0.1179$ solar radii, and $\text{albedo} = 0.30$.

Planet	P (days)	a (AU)	v (km/s)	Flux (S_E)	T_eq (K)	HZ
b	1.5109	0.01154	83.09	4.153	361.7	Hot
c	2.4218	0.01581	71.00	2.214	309.1	Hot
d	3.9275	0.02182	60.43	1.162	263.1	Opt.
e	5.8906	0.02859	52.79	0.677	229.8	Cons.
f	9.8145	0.04018	44.53	0.343	193.9	Opt.
g	11.6641	0.04507	42.04	0.272	183.0	Opt.

Table 11. Orbital and thermal properties. HZ: Cons. = Conservative HZ, Opt. = Optimistic HZ, Hot = Too hot. Flux in Earth insolation units. T_{eq} assumes $\text{albedo} = 0.30$, circular orbit, no atmosphere. Periods are TLS-detected values after Phase 8 linear ephemeris refinement.

9.3 Bulk Densities

Bulk densities were computed from pipeline-derived masses (Phase 8 TTV resonance scaling) and radii (Phase 7 batman fitting) using $\rho = M / ((4/3)\pi R^3)$. Both quantities carry documented limitations; the resulting densities range from 1.07 to 53.15 g/cm³, with planet c producing a physically impossible value (denser than osmium) due to its severely overestimated TTV mass. These results are presented as honest pipeline outputs.

Planet	M (Me)	R (Re)	Density (g/cc)	Published (g/cc)	Notes
b	12.82	1.282	8.01	~3.0	Overest.
c	63.44	1.162	53.15	~3.7	Unphysical
d	19.93	1.451	8.58	~4.2	Overest.

Planet	M (Me)	R (Re)	Density (g/cc)	Published (g/cc)	Notes
e	28.48	1.471	11.77	~4.2	Overest.
f	9.58	1.854	1.98	~3.7	Low
g	4.75	1.800	1.07	~4.4	Low

Table 12. Pipeline bulk densities compared to approximate published values from Agol et al. (2021). The scatter across orders of magnitude reflects compounding errors from both the Phase 7 radius and Phase 8 mass measurements.

9.4 Earth Similarity Index

The Earth Similarity Index (ESI; Schulze-Makuch et al. 2011, *Astrobiology* 11, 1041) combines radius, bulk density, escape velocity, and surface temperature using a weighted geometric mean:

ESI = product over i of: $(1 - |x_i - x_{i_earth}| / (x_i + x_{i_earth}))^{(w_i / n)}$

Reference values: $R_{earth} = 1.0 R_e$ ($w=0.57$), $\rho_{earth} = 5.51 \text{ g/cm}^3$ ($w=1.07$), $v_{esc} = 11.186 \text{ km/s}$ ($w=0.70$), $T_{surf} = 288 \text{ K}$ ($w=5.58$); $n = 4$.

Planet	ESI radius	ESI density	ESI velocity	ESI temp	ESI total	Rank
d	0.891	0.769	0.549	0.773	0.291	1
b	0.928	0.803	0.599	0.511	0.228	2
e	0.886	0.618	0.499	0.514	0.141	3
f	0.817	0.505	0.708	0.297	0.087	4
g	0.825	0.301	0.827	0.245	0.050	5
c	0.957	0.167	0.367	0.818	0.048	6

Table 13. Earth Similarity Index components and totals for all six planets. All scores below 0.30 reflect compounding pipeline limitations. For context, Mars scores ~0.64 and Venus ~0.44.

Phase 10 — Final Summary

Notebook: [Open in Colab](#)

10.1 Habitable Zone Classification

Habitable zone boundaries were taken from Kopparapu et al. (2013, ApJ 765, 131). The conservative HZ spans 0.36 to 1.08 Earth insolation flux; the optimistic HZ spans 0.20 to 1.78 Earth flux.

- Planet b at 4.153 S_E : Too hot — firmly outside the optimistic HZ inner edge
- Planet c at 2.214 S_E : Too hot — above the optimistic HZ inner edge of 1.78 S_E
- Planet d at 1.162 S_E : Optimistic HZ inner region ($1.08 < \text{flux} \leq 1.78$). Note: published literature classifies d as too hot based on more precise orbital parameters; the pipeline classification reflects the 3.7% period offset.
- Planet e at 0.677 S_E : Conservative HZ (0.36 to 1.08 S_E) — best habitability candidate.
- Planet f at 0.343 S_E : Optimistic HZ outer region. Published literature classifies f as conservative HZ; discrepancy arises from the 6.6% period offset increasing the calculated semi-major axis.
- Planet g at 0.272 S_E : Optimistic HZ outer region. Consistent with published classification.

10.2 Scientific Interpretation

Planet e is the strongest habitability candidate in the TRAPPIST-1 system from this pipeline's analysis. Its conservative HZ placement at 0.677 Earth flux, combined with its equilibrium temperature of 229.8 K, is consistent with the potential for liquid surface water under plausible atmospheric conditions. The Agol et al. (2021) measurements give planet e a bulk density of approximately 1.02 times that of Earth — consistent with a rocky composition with moderate volatile content. JWST observations of planet e are a high scientific priority.

The bare-rock equilibrium temperatures for the HZ planets look cold because M-dwarf luminosity is very low. The relevant quantity for habitability is the stellar flux: by flux all three of TRAPPIST-1's inner HZ planets receive amounts comparable to Mars (0.43 S_E) to Earth (1.0 S_E).

10.3 Final Pipeline Summary Table

Pl.	P (d)	R _p (R _E)	M (M _E)	a (AU)	Flux	T _{eq} (K)	ESI	HZ Status
b	1.5109	1.282	12.82	0.0115	4.153	361.7	0.228	Too hot
c	2.4218	1.162	63.44	0.0158	2.214	309.1	0.048	Too hot
d	3.9275	1.451	19.93	0.0218	1.162	263.1	0.291	Opt. HZ inner
e	5.8906	1.471	28.48	0.0286	0.677	229.8	0.141	Conservative HZ
f	9.8145	1.854	9.58	0.0402	0.343	193.9	0.087	Opt. HZ outer
g	11.6641	1.800	4.75	0.0451	0.272	183.0	0.050	Opt. HZ outer

Table 14. Final pipeline summary table. All values are pipeline-derived without substitution of published values. Masses from Phase 8 TTV resonance scaling (order-of-magnitude estimates). Radii from Phase 7 batman fitting (upper limits for d-g). Periods are TLS-detected values after Phase 8 linear ephemeris refinement.

Pipeline Limitations and Future Work

Fundamental Limitations of the 79-Day Baseline

- Period precision for outer planets (d-g): TLS targeted searches recovered periods 3-7% offset from published values. At 12-18 transits per outer planet, the periodogram cannot distinguish between the true period and aliases within a few percent.
- Radius overestimation: The 3-7% period offsets propagate into 50-80% radius overestimation in Phase 7, because transit windows are misaligned relative to the true transit centres. Batman radii for d-g are treated as upper limits throughout.
- TTV super-period recovery: The b-c TTV super-period (~757 days) is nearly ten times the K2 baseline. The d-e super-period (~36 days) is the only pair where the K2 baseline covers a significant fraction of one TTV cycle.
- Mass estimation accuracy: First-order resonance scaling masses are overestimated by 10-50x compared to published values. These are treated as order-of-magnitude upper limits.
- Outer planet detection required priors: Planets e, f, g could not be recovered in blind Phase 5 searches at the 79-day baseline. Published period windows were required.

Centroid Diagnostic Limitation

The MOM_CENTR PSF centroid columns in the K2 pipeline light curve are entirely NaN for TRAPPIST-1 (identified in Phase 1). As a result, the centroid motion diagnostic (Diagnostic 4 in Phase 6) was inconclusive for all planets. The standard eclipsing binary exclusion test based on centroid offset cannot be applied to this target with K2 data alone.

Pipeline Portability

The pipeline architecture is validated for: blind detection of inner planets below 4 days from a single K2 campaign; per-chunk PLD roll correction achieving 79.8% noise reduction; individual transit timing measurement; TTV signal detection consistent with literature amplitudes; and self-consistent orbital characterisation verifiable via stellar mass recovery. Key parameters requiring reconfiguration for a new target:

- Stellar parameters: M_{star} , R_{star} , L_{star} , T_{star}
- Quality flag encoding: check instrument documentation for SAP_QUALITY bitmask values
- SG window: scale to cadence — $\text{window_in_points} = \text{window_time_days} / \text{cadence_days}$
- GP rho parameter: set to approximate stellar rotation period
- Roll correction (K2-specific): replace with equivalent systematic correction for other missions (e.g., TESS scattered light, Spitzer pixel phase effect)

Future Work

- Multi-epoch combination: combining Campaign 12 with K2 Campaign 19 and Spitzer data would extend the baseline to several years, enabling precise period recovery for all seven planets and reliable TTV mass estimation
- MCMC fitting: replacing the Nelder-Mead single-point optimiser in Phase 7 with an MCMC sampler (e.g., emcee) would provide posterior distributions and formal uncertainties for all batman parameters
- Stellar characterisation: completing a stellar rotation period recovery from a Lomb-Scargle periodogram and flare catalogue with rates and energy distributions

- JWST follow-up context: the precise periods and transit times from this pipeline provide a starting point for scheduling JWST transmission spectroscopy observations of TRAPPIST-1 planets
- CETRA 2D linear output: the TTV-ready 2D linear search output from Phase 4 could be used directly to probe individual transit timing without re-running the detection algorithm

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Appendix A — Key Equations

A.1 Kepler's Third Law

Semi-major axis from orbital period: $a^3 = (G * M_{\text{star}} / 4\pi^2) * P^2$

where $G = 6.674e-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, M_{star} in kg, P in seconds, a in metres. [Phase 9]

A.2 Transit Depth and Planet Radius

Transit depth = $(R_{\text{planet}} / R_{\text{star}})^2 = \delta_F / F$

$R_{\text{planet}} = R_{\text{star}} * \sqrt{\text{transit_depth}} = R_{\text{star}} * r_p$ [Phase 7]

A.3 CETRA Likelihood Ratio (Linear Search)

$LR(i,j) = \text{sum over transit window of: } [-(\Delta_{ij} * m_n - f_n)^2 / (2\sigma_n^2)] \text{ minus sum of: } [-f_n^2 / (2\sigma_n^2)]$

where Δ_{ij} is the maximum-likelihood transit depth, m_n is the transit model element at position n , f_n is the observed flux, and σ_n is the flux uncertainty. [Phase 4, Smith et al. 2025, eq. 2]

A.4 Equilibrium Temperature

$T_{\text{eq}} = T_{\text{star}} * \sqrt{R_{\text{star}} / (2a)} * (1 - \text{albedo})^{1/4}$ [Phase 9]

A.5 Stellar Flux in Earth Insolation Units

$S/S_{\text{earth}} = (L_{\text{star}}/L_{\text{sun}}) * (1\text{AU}/a)^2$ [Phase 9]

A.6 Orbital Velocity (Circular Orbit)

$v = 2\pi a / P$ [Phase 9]

A.7 Earth Similarity Index

ESI = product over $i=1$ to n of: $(1 - |x_i - x_{i0}| / (x_i + x_{i0}))^{(w_i / n)}$ [Phase 9]

Reference: $R_{\text{earth}} = 1.0 R_e$ ($w=0.57$), $\rho = 5.51 \text{ g/cm}^3$ ($w=1.07$), $v_{\text{esc}} = 11.186 \text{ km/s}$ ($w=0.70$), $T_{\text{surf}} = 288 \text{ K}$ ($w=5.58$); $n = 4$

A.8 First-Order TTV Mass Scaling

$M_{\text{perturber}} / M_{\text{star}} \sim (\delta_t / P_{\text{planet}}) * (\Delta / j)$ [Phase 8]

where δ_t is TTV semi-amplitude in days, P_{planet} is orbital period in days, Δ is resonance offset, and j is resonance order.

End of Report